

Course: Cloud and Network Security

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Week 1: Network Standards, Models & Protocols

Assignment 1: Investigate the TCP/IP and OSI models in action

Introduction

This simulation activity involves using Cisco Packet Tracer to explore how the TCP/IP protocol suite operates in a live network environment. Using Packet Tracer's simulation mode allows for the direct visualization of data encapsulation and Protocol Data Units (PDUs) as packets move across network layers.

Objectives

Part 1: Examine HTTP web traffic

Part 2: Display elements of TCP/IP protocol suite

Requirements

1. Cisco Packet Tracer
2. PC
3. Internet Connection

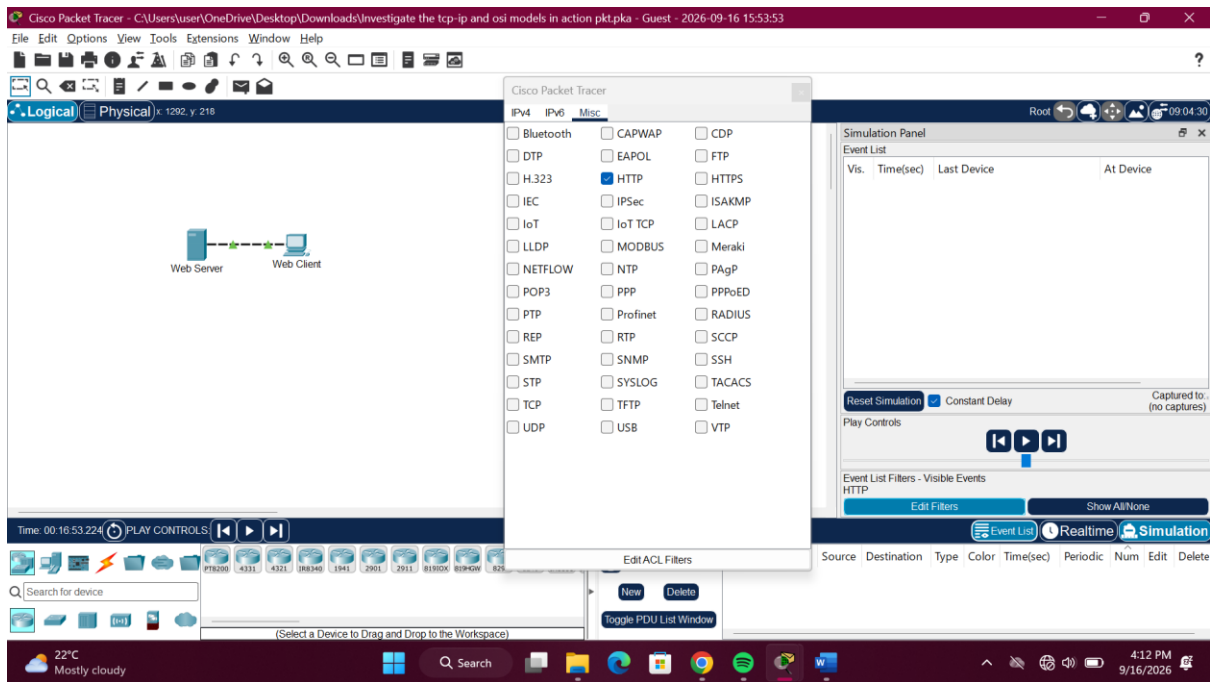
Part 1: Examine HTTP web traffic

You will use Packet Tracer (PT) Simulation mode to generate web traffic and examine HTTP.

Step 1: Switch from Realtime to Simulation mode.

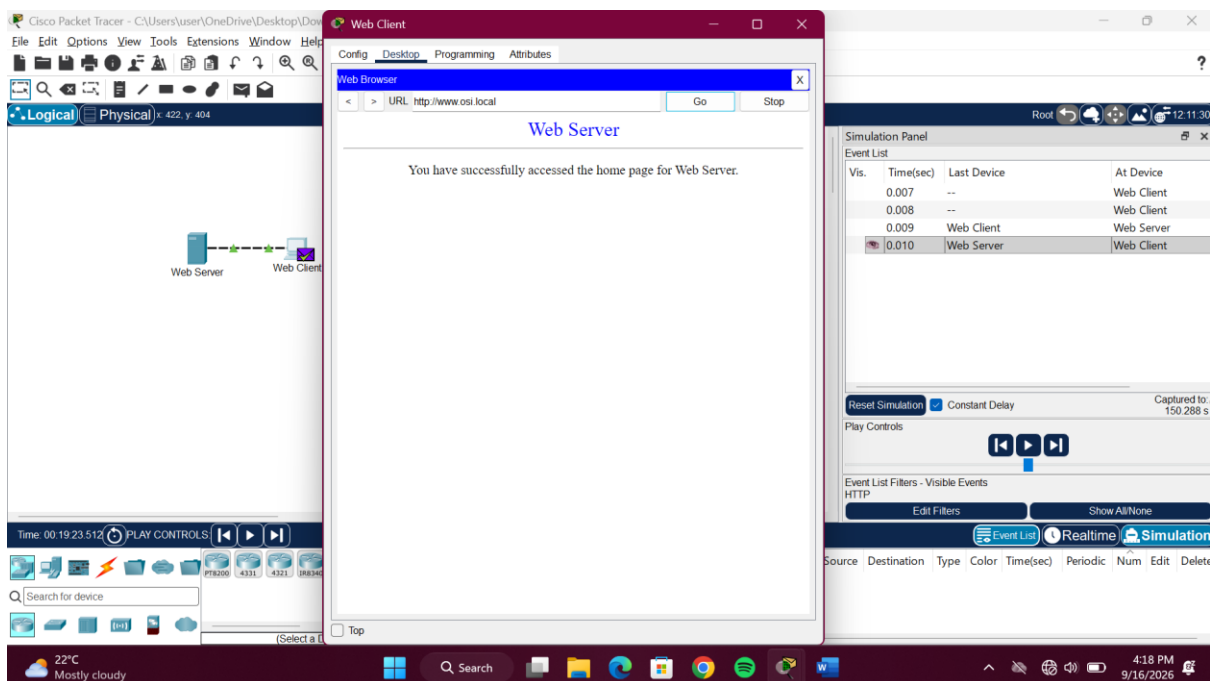
In the lower right corner of the Packet Tracer interface, click the **Simulation** mode icon to switch from Realtime mode to Simulation mode.

Select HTTP from the Event List Filters.



Step 2: Generate web (HTTP) traffic

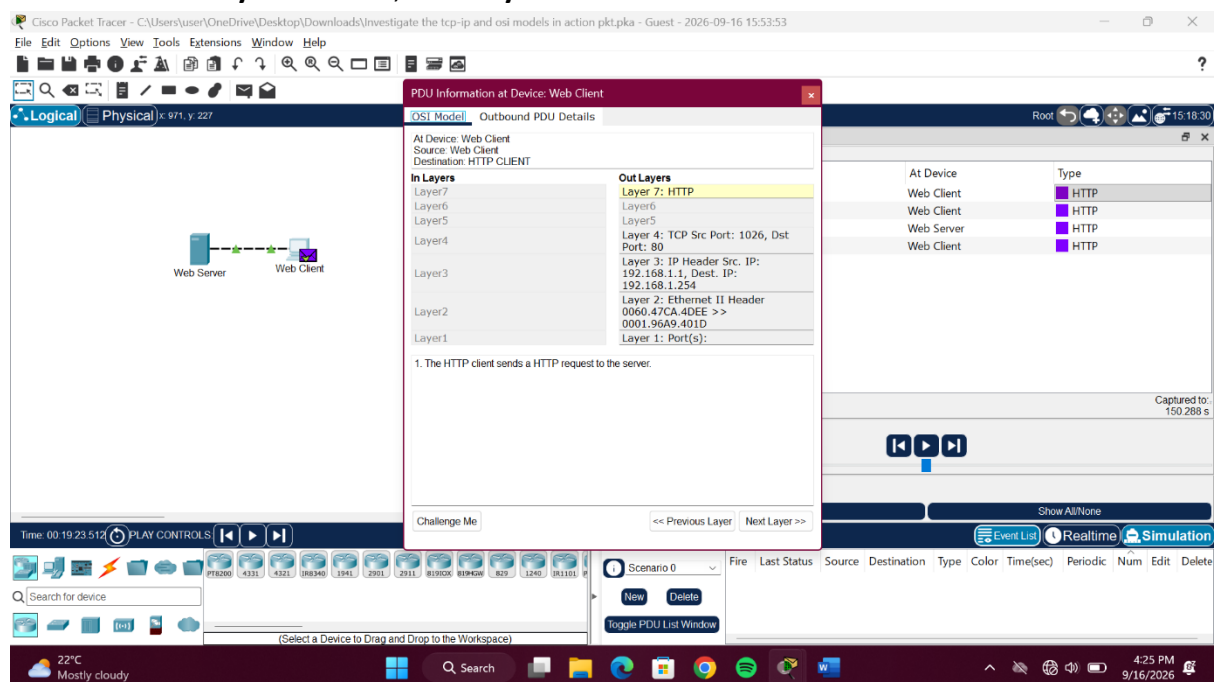
- Click Web Client in the far left pane.
- Click the Desktop tab and click the Web Browser icon to open it.
- In the URL field, enter **www.osi.local** and click **Go**.



Observations:

- Events start listing in the Event List
- The web client web browser page displays a success message showing successful connection to the home page of the web server.
- An envelope with a tick symbol appears on the web client showing successful transfer of packets.

- Click the first colored square box under the **Event List > Type** column.
- Ensure that the **OSI Model** tab is selected.
- Under the **Out Layers** column, click **Layer 7**.

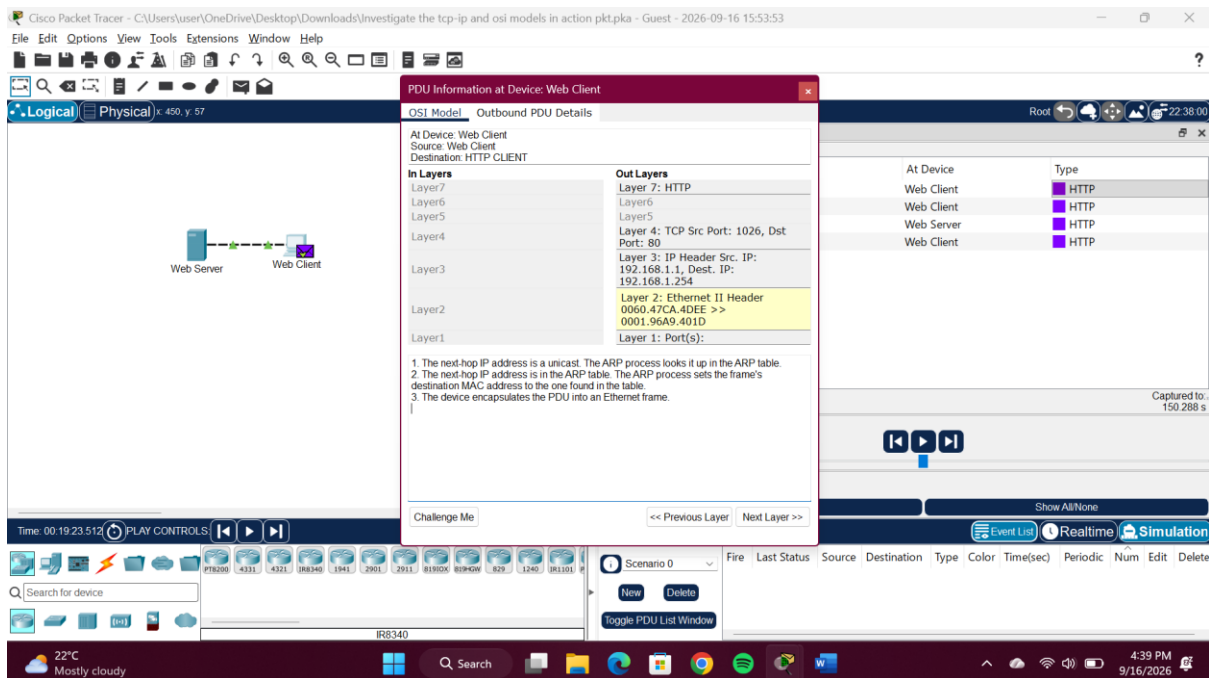


The information listed in the numbered steps is that **“The HTTP client sends a HTTP request to the server”**.

The **Dst port** value for **Layer 4** under the **Out Layers** column is **80**.

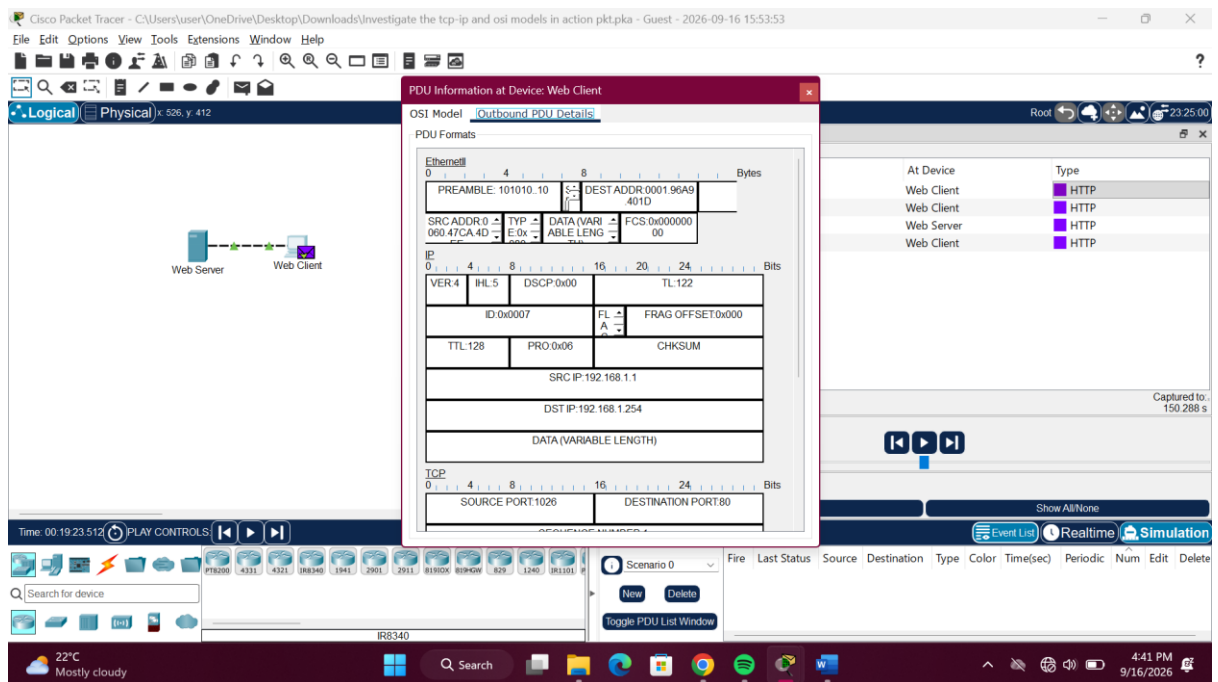
The **Dst IP** value for **Layer 3** under the **Out Layers** column is **192.168.1.254**.

The information displayed in layer 2 under the **Out Layer** column is the Ethernet II Header information.



g. Click the **Outbound PDU Details** Tab.

Information listed under the **PDU Formats** is reflective of the layers within the TCP/IP model.

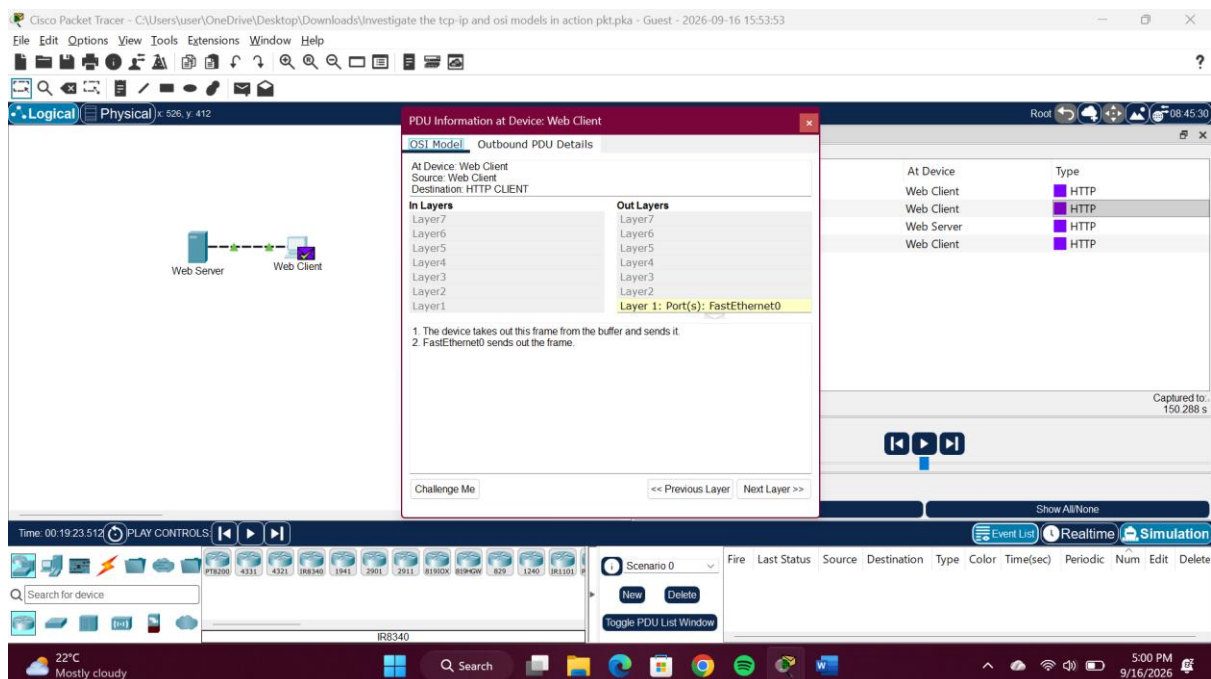


-The common information listed under the **IP** section of **PDU Details** compared to the information listed under the **OSI Model** Tab is the **Source IP address (192.168.1.1)** **Destination IP address (192.168.1.254)**. It is associated with the **third layer** of the OSI model.

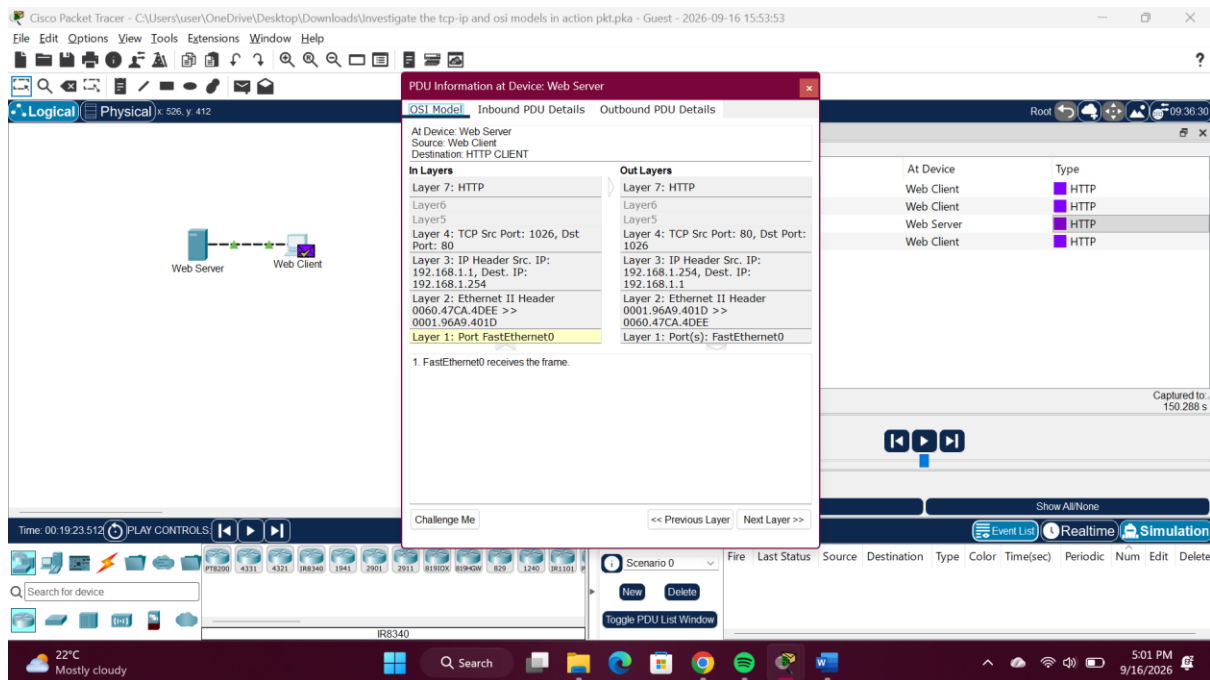
-The common information listed under the **TCP** section of **PDU Details**, as compared to the information listed under the **OSI Model** tab is the **Source Port (1026)** and **Destination Port (80)**. It is associated with the **fourth layer** of the OSI Model.

-The **Host** listed under the **HTTP** section of the **PDU Details** is **www.osi.local**. This information is associated with under the **seventh layer** in **OSI Model** tab?

- h. Click the next colored square box under the **Event List > Type** column.
Only Layer 1 is active (not grayed out). The device is moving the frame from the buffer and placing it on to the network.



- i. Advance to the next HTTP **Type** box within the **Event List** and click the colored square box.
This window contains both **In Layers** and **Out Layers**.



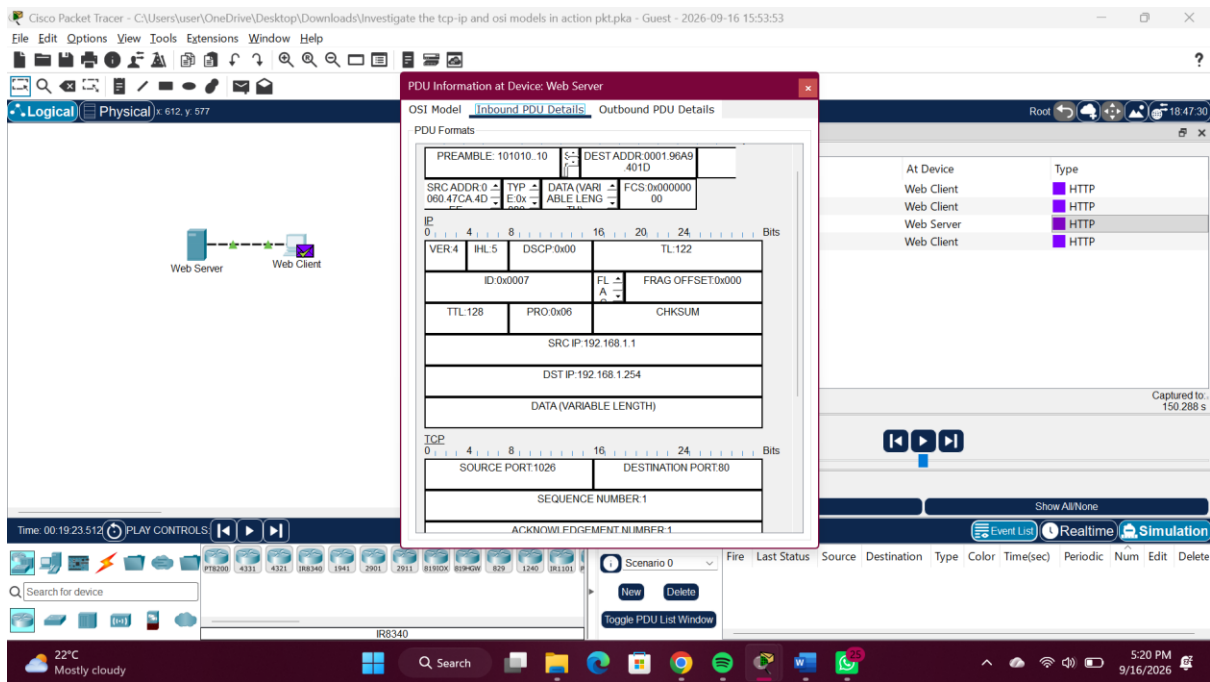
Observation: Notice the direction of the arrow directly under the **In Layers** column; it is pointing upward, indicating the direction the data is travelling.

- At the top of the column the arrow points to the right. This denotes that the server is now sending the information back to the client.

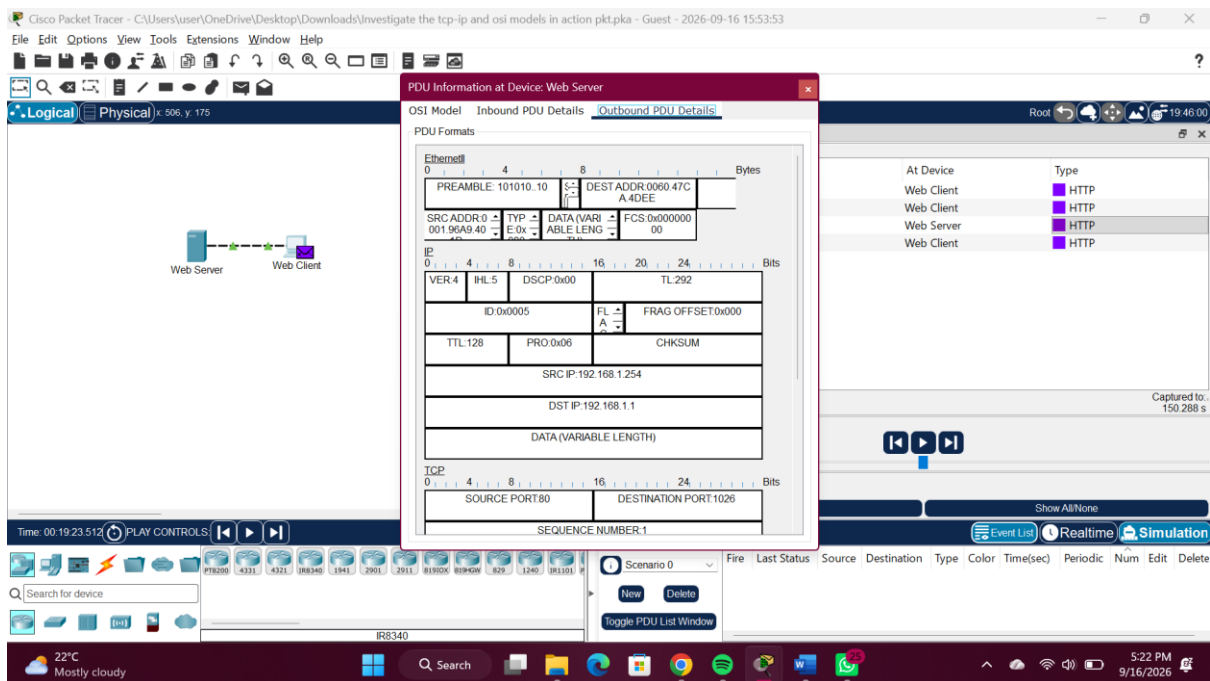
Comparing the information displayed in the **In Layers** column with that of the **Out Layers** column, these are the major differences:

- For the In Layers, the TCP Src Port is 1026 and the Dst Port is 80 while for the Out Layers, the TCP Src Port is 80 and the Dst Port is 1026.
- For the In Layers, the Header Src IP is 192.168.1.1 and the Dest IP is 192.168.1.254 while for the Out Layer, the Header Src IP is 192.168.1.254 and the Dest IP is 192.168.1.1.

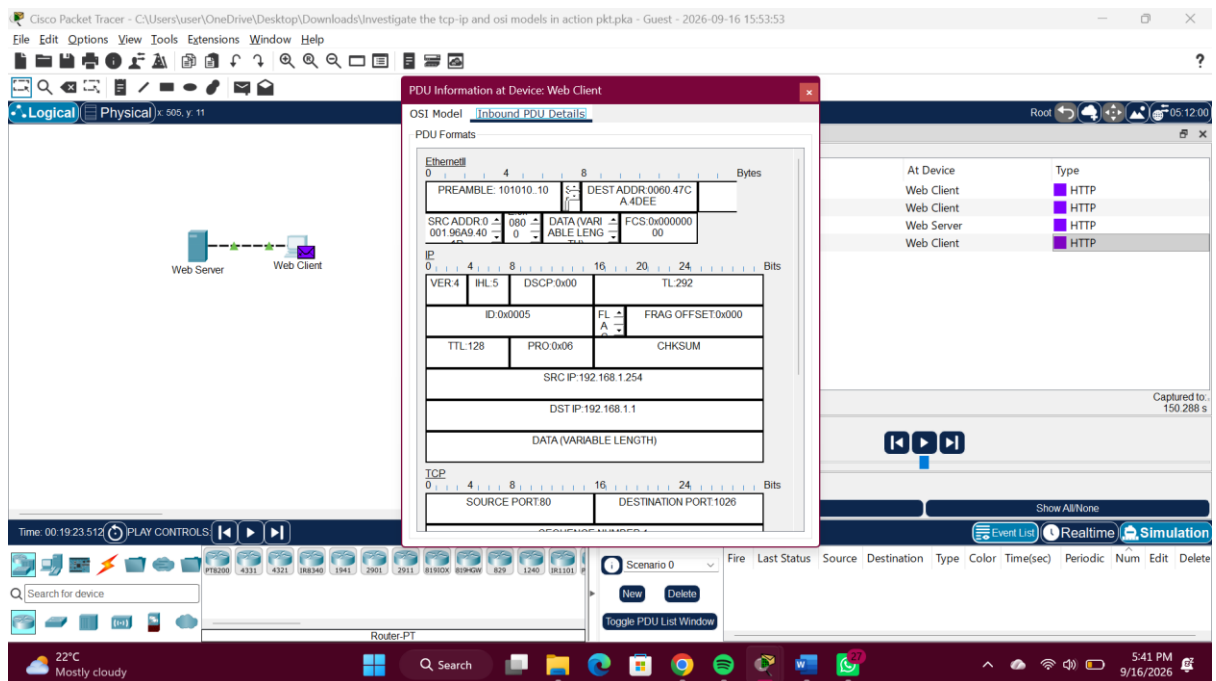
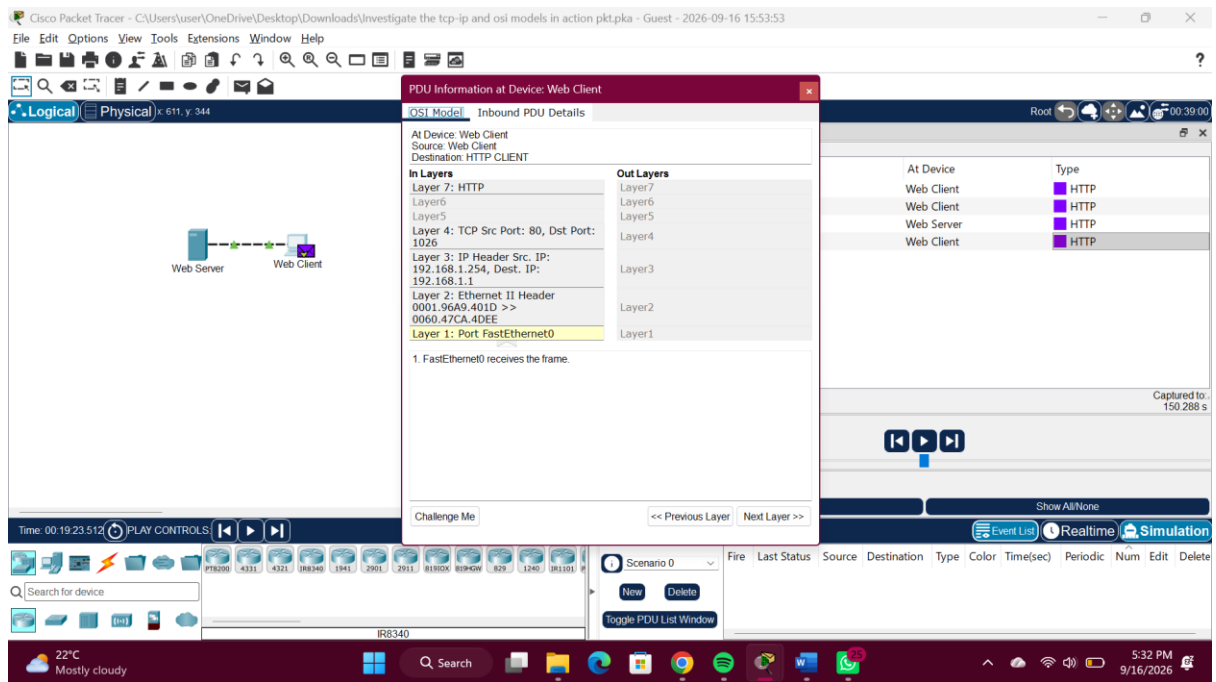
- Click the **Inbound and Outbound PDU Details** tab. Review the PDU details.
 - Inbound PDU Details



- Outbound PDU Details



- Click the last colored square box under the **Info** column.



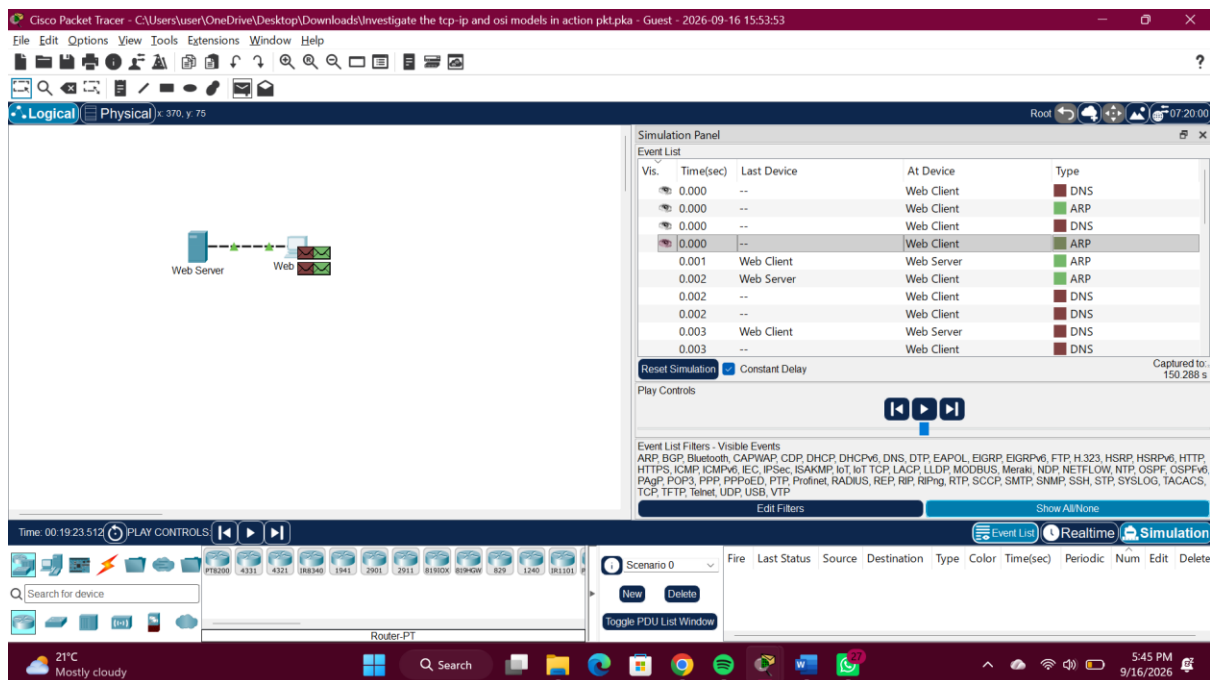
There are two tabs associated with this event, the **OSI Model** tab and the **Inbound PDU details**. The OSI Model tab provides an overview of the data being processed through the network layers while the Inbound PDU details tab contains detailed header information of the incoming frame.

Part 2: Display Elements of the TCP/IP Protocol Suite

In this part, we use Packet Tracer Simulation mode to view and examine other protocols of comprising of the TCP/IP suite.

Step 1: View Additional Events

- Close any open PDU information windows.
- In the **Event List Filters > Visible Events** section, click **Show All/None**.

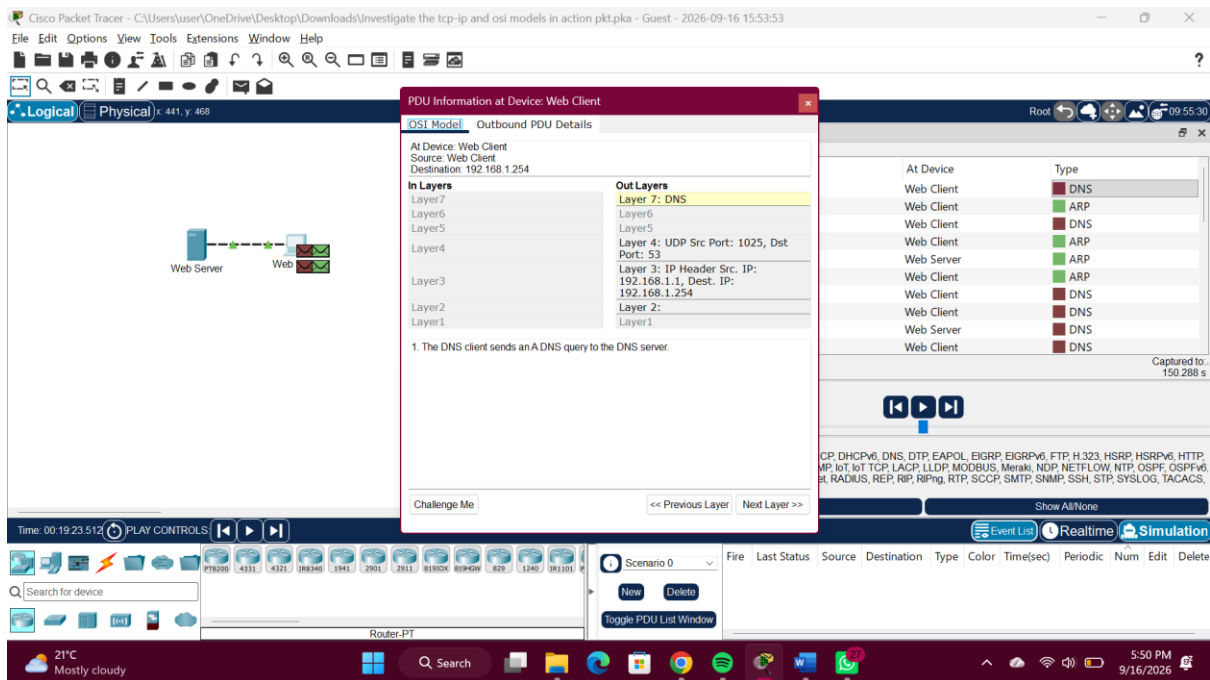


- Additional events are displayed. Also, additional entries occur in the table.
- These extra entries play various roles within the TCP/IP suite.
- Address Resolution Protocol (ARP) requests MAC addresses for destination hosts.
- DNS is responsible for converting a name (for example, **www.osi.local**) to an IP address.
- The additional TCP events are responsible for connecting, agreeing on communication parameters, and disconnecting the communications sessions between the devices.

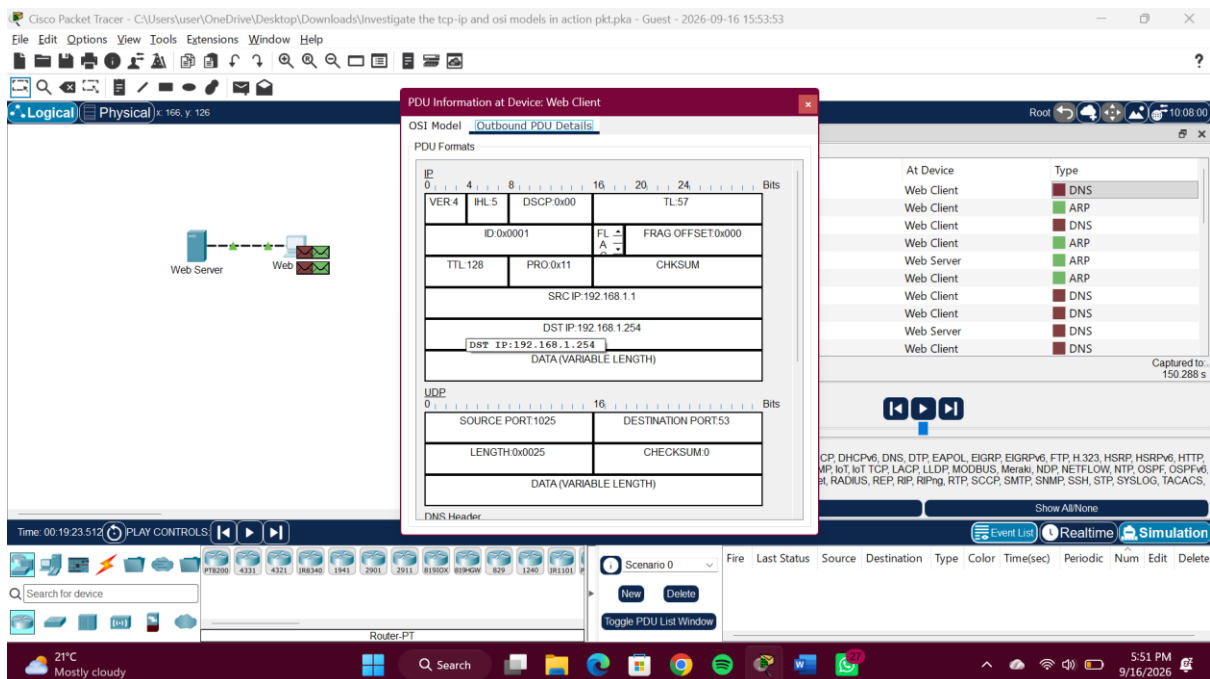
- Click the first DNS event in the **Type** column.

This opens to the OSI Model tab.

Layer 7 is highlighted and a description of what is occurring is listed directly below the **In Layers** and **Out Layers**; "The DNS client sends a DNS query to the DNS server."

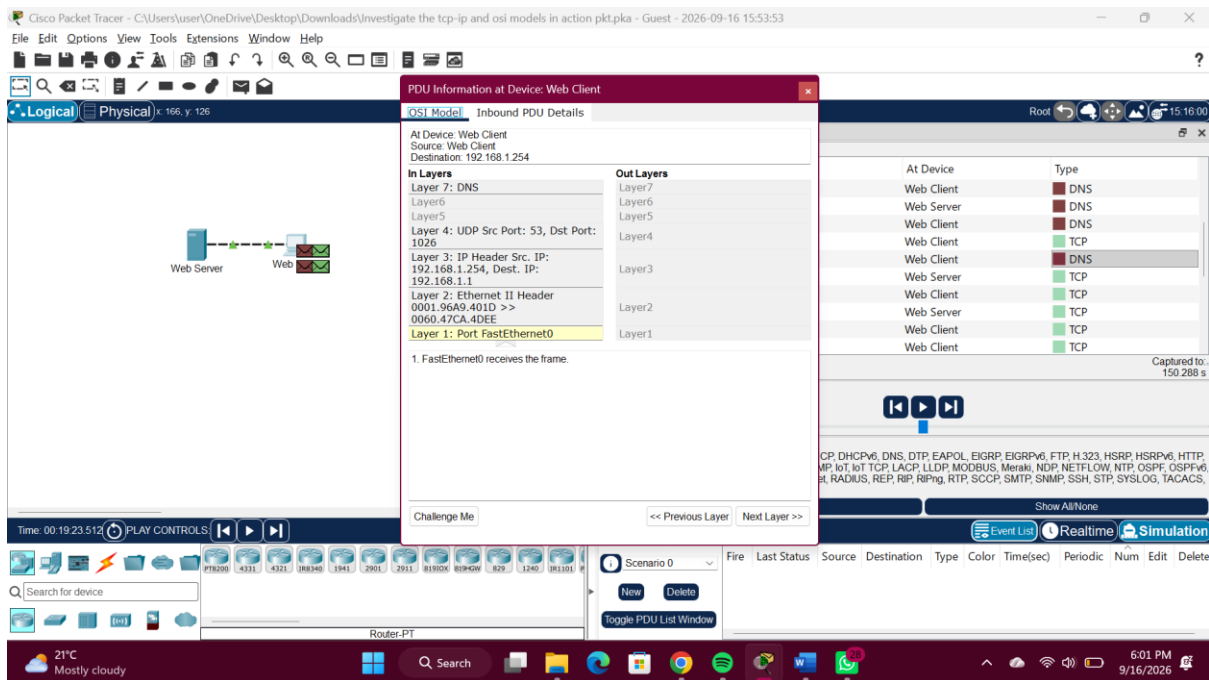


d. Click the **Outbound PDU Details** tab.

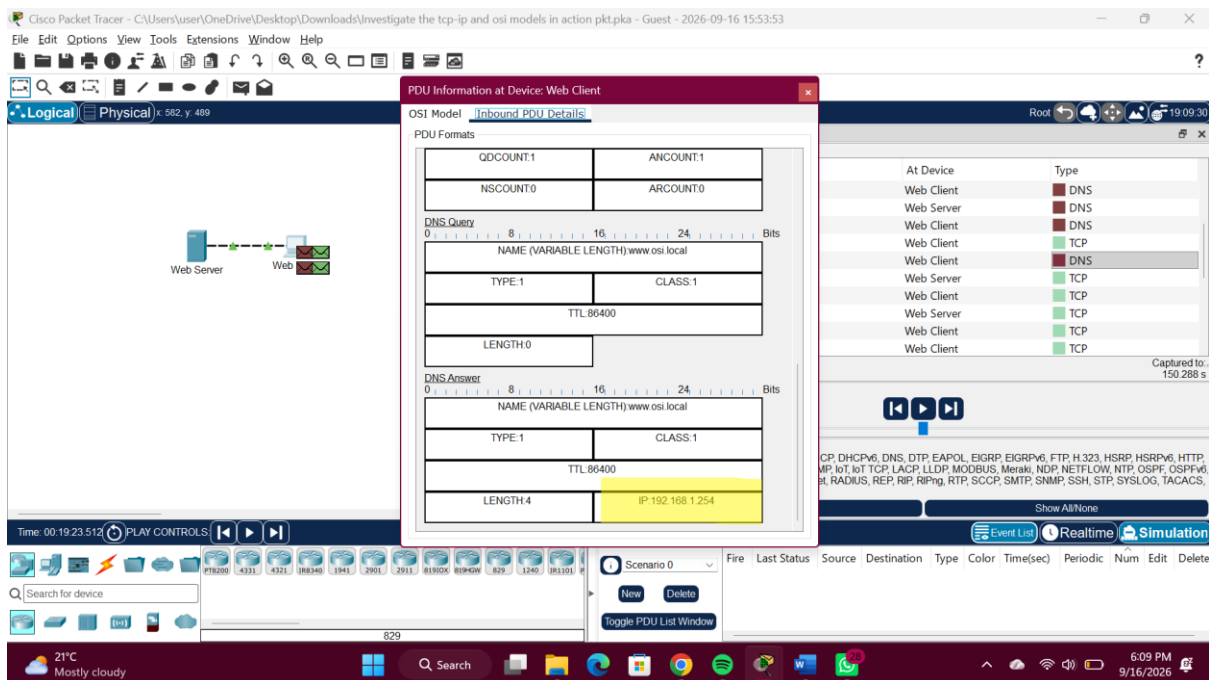


The information is listed in the **NAME** field: in the DNS QUERY section is www.osi.local.

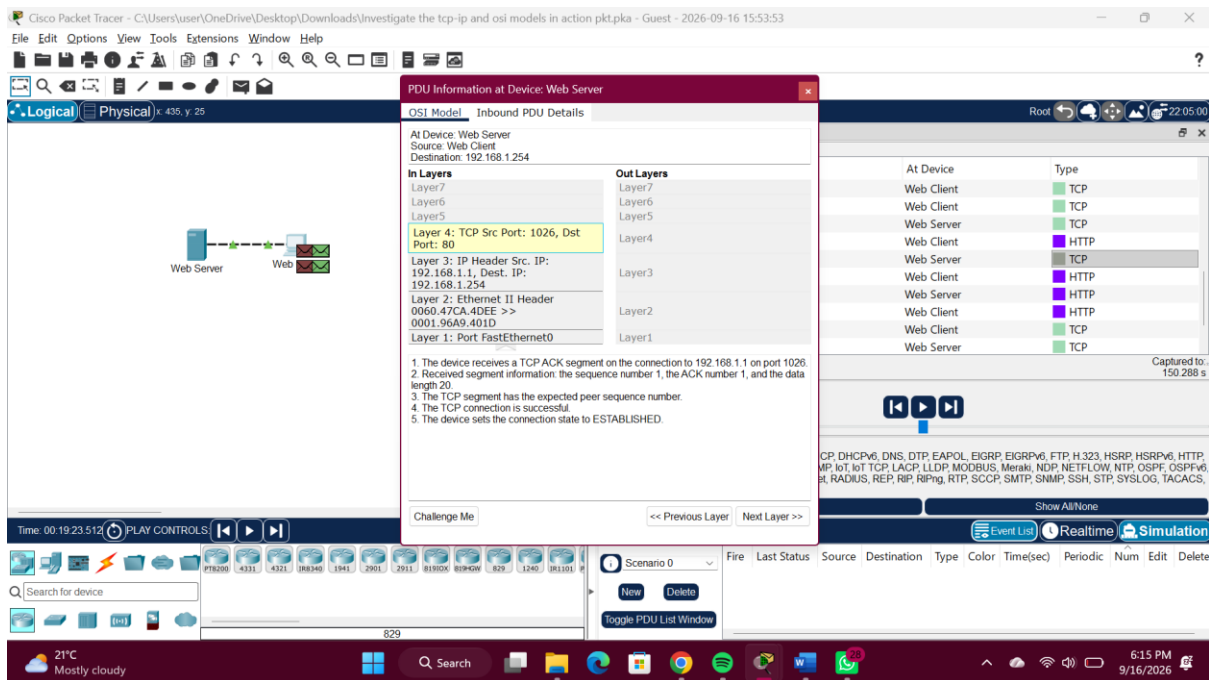
e. Click the last DNS Info colored square box in the event list.



- The PDU was captured from the **Web Client**.
- The value listed next to **ADDRESS:** in the DNS ANSWER section of the **Inbound PDU Details** is 192.168.1.254.



- Find the first **HTTP** event in the list and click the colored square box of the **TCP** event immediately following this event. Highlight **Layer 4** in the **OSI Model** tab.



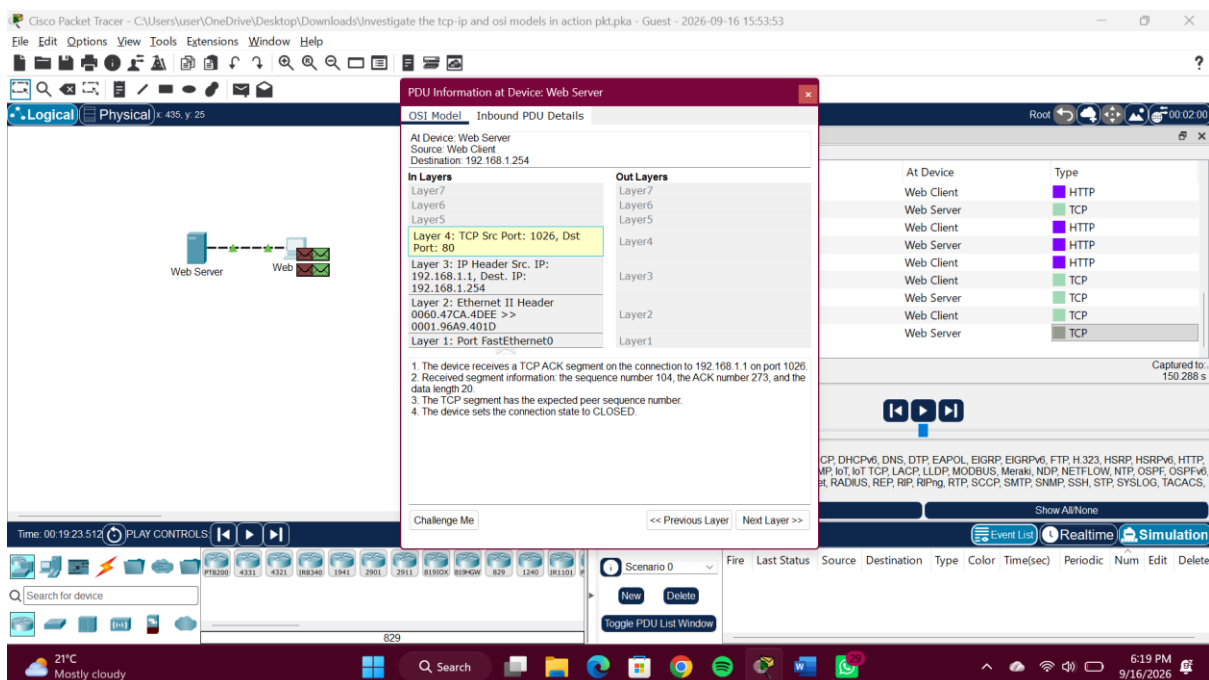
In the numbered list directly below the **In Layers** and **Out Layers**, the information displayed under items 4 and 5 are:

4. The TCP connection is successful.
5. The device sets the connection state to ESTABLISHED.

g. Click the last TCP event.

Highlight Layer 4 in the **OSI Model** tab.

Examine the steps listed directly below **In Layers** and **Out Layers**.



Based on the steps listed in Layer 4, the purpose of this event to formally terminate the TCP connection between the Web Server and the Web Client.

Conclusion

Based on the information that was inspected during the Packet Tracer capture, the Web Server is listening on PORT 80 for the web requests and on Port 53 for DNS requests.